MATERIAL	SPEC NUMBER	REVISION	EFFECTIVE DATE
17-4 PH, 105K, NACE	CMS-303	H	10/18/2024

Material Requirement

Scope	17-4PH (ASTM A564 GR 630 or UNS S17400 Modified), 105K Minimum Yield Strength, Stainless Steel, NACE MR0175/ISO 15156
Product Forms	Bar and Mill Shapes, Ring Forgings, Open and Closed Die Forgings
	CMS-002, Material Requirements for Forgings, Bars and Mill Shapes
General Requirements	Many values are listed as inch-pound units and SI(metric) units and are to be regarded separately. The values stated in each system are not exact equivalents; therefore, each system must be used independent of the other. All material shall be capable of passing API 6A PSL3 requirement for volumetric, surface NDE, and charpy impacts at -75°F (-60°C)

Chemical Composition

Element	Composition Limits	Element	Composition Limits
Carbon	0.07 Maximum	Manganese	1.0 Maximum
Silicon	1.00 Maximum	Phosphorous	0.040 Maximum
Chromium	15.00 – 17.00	Copper	3.00 – 5.00
Sulfur	0.030 Maximum	Niobium	0.15 – 0.45
Nickel	3.00 – 5.00	Iron	Balance

NOTE: Niobium and Columbium refer to the same element. Additional elements that can be readily tested shall be included in the final report.

Special Requirements for Cactus Forgings

Modified chemistry for the below elements. Those elements not listed below shall comply with the above limits.

Chromium: 15.00 – 16.50 **Copper**: 3.00 – 4.50 **Nickel**: 4.00 – 5.00

Delta Ferrite should be 3% max. through one field of view (Determining volume fraction by systematic manual point count)

Mechanical Properties

Properties	Range
Tensile Strength, psi	125,000 (860 MPa) Minimum
Yield Strength, psi, 0.2% Offset	105,000 (725 MPa) Minimum
Elongation in 2" Gage Length	16% Minimum
Reduction in Area	50% Minimum
Brinell Hardness (HBW)	277 - 311 HBW
Rockwell "C" Hardness (HRC)	29 – 33 HRC

Melt Practice	Electric furnace (EF) or vacuum induction melting (VIM) process, ladle refined (LF), Argon Oxygen Decarburization (AOD). Primary melting process may be followed by secondary melting using electroslag (ESR) or vacuum arc melting (VAM). If secondary remelting is used, a heat shall be defined as all the ingots remelted from a single primary heat.
Condition	Solution anneal and double age harden per one of the options below:

First Option - Double Age Heat Treatment (Double H1150)			
	Atmos/Media	**Temperature**	**Time at Temp**
Heat Treatment Solution Anneal	Air or Nitrogen	1875°F - 1925°F (1024°C-1052°C)	1 Hour Minimum
	Air Cool or Liquid Quench to Below 90°F (32°C). Allow enough time for cooling so the entire section has cooled to the required temperature. The core of the material shall be at temp for at least 30 minutes.		
Heat Treatment First Age	Air or Nitrogen	1125°F - 1175°F (607°C - 635°C)	4 Hours Minimum
	Air Cool or Liquid Quench to Below 90°F (32°C). Allow enough time for cooling so the entire section has cooled to the required temperature. The core of the material shall be at temp for at least 30 minutes.		
Heat Treatment Second Age	Air or Nitrogen	1125°F - 1175°F (607°C - 635°C)	4 Hours Minimum
	Air Cool or Liquid Quench to Below 90°F (32°C).		
Second Option – Double Age Heat Treatment (H1150M)			
	Atmos/Media	**Temperature**	**Time at Temp**
Heat Treatment Solution Anneal	Air or Nitrogen	1875°F - 1925°F (1024°C - 1052°C)	1 Hour Minimum
	Air Cool or Liquid Quench to Below 90°F (32°C). Allow enough time for cooling so the entire section has cooled to the required temperature. The core of the material shall be at temp for at least 30 minutes.		
Heat Treatment First Age	Air or Nitrogen	1375°F - 1425°F (746°C - 774°C)	2 Hours Minimum
	Air Cool or Liquid Quench to Below 90°F (32°C). Allow enough time for cooling so the entire section has cooled to the required temperature. The core of the material shall be at temp for at least 30 minutes.		
Heat Treatment Second Age	Air or Nitrogen	1125°F - 1175°F (607°C - 635°C)	4 Hours Minimum
	Air Cool or Liquid Quench to Below 90°F (32°C).		
	Note: The H1150M heat treatment may result in a hardness as low as 26 HRC.		

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Date Prepared	*October 18, 2024*	***Date Approved:***	*October 18, 2024*

Revision History Table		
Rev H	10/18/2024	Added additional requirements for China Closed/Open die forgings.
Rev G	05/26/2021	Align with NACE MR0175/ISO 15156-2020.
Rev F	09/23/2020	Align with ASTM A564/A564M-19.
Rev E	07/17/2020	Revised Columbium+Tantalum, to Niobium, length of H1150M first age heat treatment, align with ASTM A564/A564M-19.
Rev D	08/30/2018	Updated brinell hardness range 277 -311 BHN.
Rev C	07/21/2012	Revised to include liquid quench as an acceptable media for second age.
Rev B	11/9/2011	Editorial changes, scope, hardness, melt condition, heat treat continuous furnace. To enhance current material specification and address accepted industry requirements.
Rev A	9/22/2011	Initial Release